LÉHO DEHOVE

BIO-MECHATRONICS · ROBOTICS · EMBEDDED SYSTEMS · COMPUTER VISION

M.Sc. Engineering Student at ESME Paris (Graduation Sept. 2027) | **Seeking 6-Month R&D Internship from April 2027**

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TECHNICAL PORTFOLIO

Verified Engineering Experiences, Hardware Prototypes & Systems

Professional Engineering Experiences

CNRS @ ESPCI PSL — **Computer Vision R&D Intern** – Brain Plasticity Lab — **June – expected Sept. 2026**

- Developed **KiAnSo**, an automated video processing software replacing legacy tools for Drosophila behavioral neuroscience. Increased analysis throughput by **+500%** and saved over **45 hours per experimental batch**.
- Engineered **OptoStim**, an automated photostimulation hardware bench and high-speed synchronized video acquisition platform.
- Co-authoring a research publication on automated neural locomotion kinematics under optical stimuli.

Danfoss Power Solutions — **Manufacturing & Quality Operator** – Fluid Power Systems Plant — **June – July 2025**

- Quality control, precision metrology, mechanical assembly, and packaging of hydraulic components for fluid power systems.

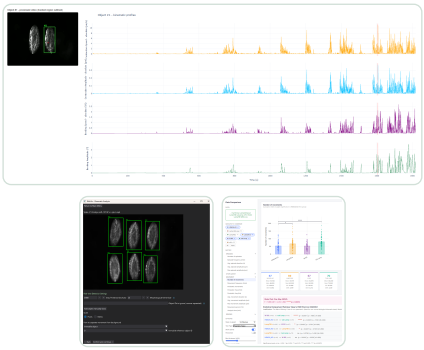
ESME Club & FabLab — **FabLab Administrator & Club Vice-President** — **Sept. 2022 – June 2025**

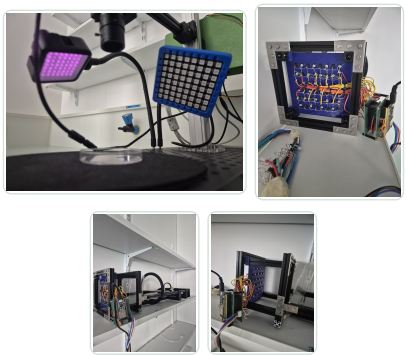
- Supervised digital fabrication equipment (CNC milling, SLA/FDM 3D printing, laser cutters) and trained **150+ engineering students** in CAD/CAM, Design for Additive Manufacturing (DFAM), and rapid physical prototyping.

Univ. Putra Malaysia (UPM) — **Academic Exchange** – Electrical & Electronic Engineering — **2025 – 2026**

- International IEEE final-year coursework in control systems, sensor fusion, and microelectronics. English proficiency C1.

Flagship Projects: Computer Vision & Neuro-Robotics Systems





KiAnSo — Kinematic Analysis Software · CNRS / ESPCI-PSL — Sept. 2026

`Python` `OpenCV` `Farnebäck Optical Flow` `Savitzky-Golay / Hampel` `Batch Processing`

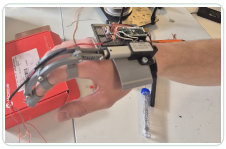
- **Scientific Problem:** In vivo behavioral neuroscience on Drosophila relied on manual, subjective frame-by-frame analysis, taking up to 5 days per dataset and introducing human annotator variability.
- **Software Architecture:** Engineered from scratch in Python. Implemented Farnebäck dense optical flow, robust signal filtering (Hampel outlier rejection and Savitzky-Golay polynomial smoothing), and automated ROI segmentation to model embryonic spasms and adult locomotion kinematics.
- **Performance Metric:** Achieved a **+500% throughput increase**, turning a 5-day week-long task into a single day and saving **over 45 hours per experimental batch**. Deployed in active lab research.

OptoStim — Optogenetics Rig & Synchronized Acquisition · CNRS / ESPCI — Sept. 2026

`Python` `Flask` `Custom LED Matrix` `Extruded Aluminum` `Microcontrollers`

- **System Objective:** Provide millisecond-accurate optical photostimulation coupled with synchronized high-speed video acquisition for phototactic behavioral neural recording.
- **Hardware & Firmware:** Designed and assembled an extruded aluminum modular rig, custom LED driver board, and embedded microcontroller firmware delivering programmable pulse trains.
- **Software Integration:** Python/Flask acquisition server coordinating camera triggers with optical excitation pulses. Directly coupled with KiAnSo for immediate post-stimulus kinematic tracking.

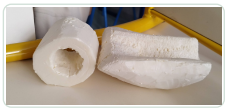
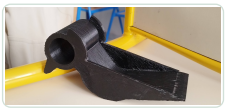
Bio-Mechatronics, Exoskeletons & Physical Robotics




AILAPS — Active Motorized Hand Exoskeleton for Rehabilitation — May 2026

Biomechatronics | SolidWorks | Embedded C | Motor Actuation | Additive Mfg

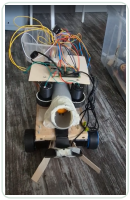
- **Clinical Purpose:** Wearable motorized orthosis to assist patients with muscular weakness and facilitate physical rehabilitation during active finger flexion/extension cycles.
- **Kinematic Design:** Modeled and simulated custom compliant multi-bar linkages in SolidWorks to match anatomical finger trajectories without mechanical joint hyperextension.
- **Control & Electronics:** Embedded deterministic C control on microcontrollers, integrated power driver stages, and fabricated ergonomic structural components via additive manufacturing.

Cuicui — Pediatric Upper-Limb Mechanical Prosthesis — June 2023

Biomechanics | Ergonomics | Medical Casting | Passive Mechanics | 3D Printing

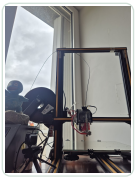
- **Real-World Case:** Custom mechanical prosthetic device engineered for a young child with congenital upper limb agenesis wishing to ride a bicycle autonomously.
- **Anatomical Socket Casting:** Cast direct limb molds using medical alginate and dental plaster to guarantee a perfectly customized socket contour without painful skin pressure points.
- **Mechanism:** Designed a passive, robust handlebar locking and rapid-release safety mechanism, providing stable bicycle steering control with zero electrical dependency or added battery weight.




Scotchouille — Autonomous Competition Mobile Rover (Tournament Winner) — June 2025

Python | OpenCV | Differential Drive | Trajectory Planning | Launcher System

- **Robotics Tournament:** Autonomous mobile robot designed to search, track, catch, and shoot targets on an arena under competition constraints. **Won 1st place in tournament**.
- **Computer Vision & Navigation:** Integrated monocular computer vision in OpenCV for rapid target detection; real-time trajectory calculation on a custom differential drive platform.
- **High-Velocity Ballistics:** Designed a high-speed flywheel launching system with servo feed control and dedicated power distribution.



Louya — Custom Direct-Drive Rapid Prototyping Bench — 2016 – Present

Additive Manufacturing | Direct-Drive Extrusion | Firmware Optimization | DFM/DFA

- Custom-modified Creality CR-10 testbed upgraded over 8+ years: custom direct-drive extrusion, all-metal hotend, precision mechanical alignments, and tuned Klipper/Marlin firmware.
- Serves as active manufacturing testbed for technical polymers (PLA, PETG, TPU, Nylon, Carbon-Fiber composites) and Design for Additive Manufacturing (DFAM) evaluation.

Control Theory & State Estimation Benchmarks

Hybrid Neural vs PID Control (Quadrotor Payload)

Control Theory | MLP in C | Cortex-M4 | Nonlinear Damping

- Tackled violent swing in suspended quadrotor payloads during aggressive maneuvers ($> 20^\circ$ tilt) where classical linear PID gains saturate.
- Trained a compact 2-layer MLP (106 parameters, 8 linear algebra equations in C, **evaluates in 2.5 μs** on ARM Cortex-M4).
- **Benchmark Result:** Hybrid architecture (PID integral baseline + neural policy) **cut tracking error by 43%** (from 27.3 cm to 15.5 cm) while rejecting steady-state wind disturbances.

3D State Estimation with Orocos BFL

C++ | Orocos BFL | EKF | Particle Filter | OpenCV

- Benchmark of Bayesian estimators on rapid deterministic 3D trajectories (4D state: $[z, \dot{z}, \theta, \omega]$ at 6 Hz spin) using OpenCV and Orocos BFL.
- **EKF:** Executed in **0.04 ms/step**, converged onto 37.7 rad/s spin in 5 frames (**96.4% prediction accuracy** in 200 ms).
- **Particle Filter (SMC):** 4.2 ms/step, maintained multi-modal probability distributions through visual occlusions and verified filter confidence drops under ill-conditioned camera angles.